

An independent audit of the September 2026 fluid blowup claims, and a trade-off between force regularity and dissipation

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Abstract

In September 2026 two groups announced finite-time blowup results for fluid equations with machine-checked certificates: Alpegé and Buckmaster for the two-dimensional inviscid Boussinesq system and three-dimensional Euler with smooth forcing, and OpenAI for three-dimensional Navier-Stokes with forcing at every positive viscosity together with unforced Euler. Both extend the multiscale layer program of Córdoba and Martínez-Zoroa. This paper reports an independent audit and four quantitative consequences, none of which is a claim about the correctness of the mathematics beyond what a machine check establishes.

First, the Lean certificate of the OpenAI development was rebuilt and then replayed through the Lean kernel from an empty environment, both halves, at the toolchain version the development pins. Second, the reduced amplitude system of the layer mechanism was transcribed, extended with fractional dissipation, and validated against direct pseudospectral simulation of the nonlinear equations, including the growth, steering and holding cycle that makes the cascade iterable. Third, the published hypodissipative threshold $(22 - 8\sqrt{7})/9$ of Córdoba, Martínez-Zoroa and Zheng is recovered exactly from their own exponent budget, with the binding constraint identified and with the consequence that the admissible frequency ratios form an interval closing precisely at that threshold, so that geometric cascades are excluded at every positive dissipation. Fourth, applying the dissipative reduction to the published smooth-forcing schedule gives a one-line trade-off $\alpha < \delta/(4Q)$ between the amplitude margin, the frequency ratio and the dissipation exponent, from which we record a dated and falsifiable prediction about the unreleased hypodissipative sequel.

Each statement is labelled machine-verified [MV], derived [D], or conjectural [C]. Two claims from an earlier round of this work are retracted in place and the retractions are kept.

1 Scope, and what is not claimed

The Clay Millennium problem asks for one of four alternatives in Fefferman’s official statement; the forced alternatives (C) and (D) are among them. Nothing in this paper bears on whether any announced proof is correct as mathematics. What is offered is an audit that a third party can

run, a validated reduced model, and exponent bookkeeping whose every constant comes from a displayed formula in a published paper.

The layer mechanism is due to Diego Córdoba and Luis Martínez-Zoroa, pushed to smooth forcing by Levent Alpogé and Tristan Buckmaster, with a different mechanism used by OpenAI for the viscous case. Nothing here is our mechanism. Throughout, our dissipation convention is $-\nu(-\Delta)^\alpha$, so classical viscosity is $\alpha = 1$; Córdoba, Martínez-Zoroa and Zheng write $|\nabla|^\alpha$, so their exponent is twice ours and their threshold $(22 - 8\sqrt{7})/9 = 0.09267$ is 0.04633 in our convention.

2 Machine verification of the Lean certificate

Statement audit [MV]. The development `openai/NavierStokesAndEuler` contains 2,659 files and 641,332 lines. Its headline statements were checked clause by clause against Fefferman’s alternatives (C) and (D). The strongest single fact is not the size of the development: the reference statement is adapted from DeepMind’s Formal Conjectures, so the definitions that decide whether the formalized theorem is the intended theorem were written by a third party before the claim existed. The unforced Euler half inherits the same reference.

Build [MV]. The full development builds cleanly on an independent machine: 11,424 jobs, zero occurrences of `sorryAx`, and all four headline theorems depending on exactly `propext`, `Classical.choice` and `Quot.sound`.

Kernel replay [MV]. A successful build and a clean axiom report both read the environment that the *elaborator* produced. Since Lean v4.28.0 the kernel checker ships with the toolchain as `leanchecker`, and the development pins `leanprover/lean4:v4.34.0-rc2`, so the check is available at the pinned version with no third-party build. Replaying every constant in the closure of each root module into an *empty* kernel environment succeeds for both halves:

target	wall clock	exit
<code>NavierStokes</code>	2,972 s	0
<code>Euler</code>	1,588 s	0

at most 6.4 GB resident. The tool’s own scope statement is retained rather than inflated: it detects environment hacking, it is not an external verifier. A per-module sweep was attempted first and abandoned for a measured reason: it loads one import closure per module and was input-output bound on this machine, accumulating 144 seconds of processor time in 80 minutes.

What this does not establish. Nothing about whether the statements are the intended ones beyond the clause-by-clause audit, and nothing about the mathematics in the sense a referee means.

3 The reduced model, and what it costs to reduce

3.1 The amplitude system and its dissipative extension [D]

On a background that is affine near the origin, $u_{\text{old}} = D(t)x$ and $\theta_{\text{old}} = G(t) \cdot x$, a temperature wave and a vorticity wave sharing one phase $s = \lambda\zeta(t) \cdot x$ satisfy a closed system,

$$\dot{\zeta} = -D^T \zeta, \quad \dot{\Theta} = -\frac{J\zeta \cdot G}{\lambda|\zeta|^2} \Omega, \quad \dot{\Omega} = \lambda \zeta_1 \Theta, \quad (1)$$

exact where the envelope is one and the profile is linear. In the frozen configuration $D = 0$, $G = -Ae_2$, $\zeta = re(\varphi)$ the growth rate is $\sqrt{A} \sin \varphi$, which carries no λ : the frequency in $\dot{\Omega}$ cancels the reciprocal frequency in $\dot{\Theta}$. Frequency buys gradient, not growth, through the amplification identity $\nabla \vartheta(0) = \lambda \Theta \zeta$.

Adding $-\nu(-\Delta)^\alpha$ to both equations is diagonal on the phase, so the whole effect is one damping term per amplitude,

$$\dot{\Theta} = \dots - \nu(\lambda|\zeta|)^{2\alpha} \Theta, \quad \dot{\Omega} = \dots - \nu(\lambda|\zeta|)^{2\alpha} \Omega, \quad (2)$$

with eigenvalues $-\nu(\lambda r)^{2\alpha} \pm \sqrt{A} \sin \varphi$ and therefore the frequency cap $\lambda \lesssim A^{1/(4\alpha)} \nu^{-1/(2\alpha)}$. Equation (2) is derived here; it is the one analytic ingredient of this paper that is not transcribed.

3.2 Validation against the equations [MV]

Direct pseudospectral simulation of the nonlinear two-dimensional Boussinesq system, in double precision with two-thirds dealiasing, reproduces the reduced model: peak growth rate to 4×10^{-5} relative at two insertion angles, frequency independence to 2.7×10^{-4} across an eightfold range of λ , and the derived dissipative term of (2) tracking the measured rate to 2.6×10^{-4} absolute across $\alpha \in \{1/4, 1/2, 1\}$ and $\nu \in \{10^{-5}, 10^{-4}\}$. Three negative controls fail as required, after a first attempt was found vacuous at an angle where the corruption changed nothing.

The cascade premise, that a layer responds to the *total* accumulated low-frequency gradient rather than to the base stratification, was tested on a frozen two-scale background that is an exact steady state (measured drift 1.45×10^{-15}): the local rate follows the total gradient at correlation 0.99945 and the base alone at 0.212.

3.3 The steering cycle, realized on a torus [MV]

The mechanism is iterable only because a grown layer can be returned to rest: growth leaves both the temperature gradient the next layer needs and a shear with coefficient Ω that would keep deforming it. The construction removes the shear by rotating G and ζ *together*, which preserves $J\zeta \cdot G$ and $|\zeta|$ but moves the laboratory component ζ_1 that multiplies Θ in $\dot{\Omega}$. In normalized time the control is a smooth ramp of ζ_1 to zero followed by a short negative pulse whose amplitude is *selected* so that the vorticity amplitude lands on zero.

A rigid rotation of the background does not fit on a torus. In two dimensions it does not have to: in the co-rotating frame the Coriolis and centrifugal accelerations are gradients and are absorbed by the pressure, and the Euler acceleration shifts only the spatially uniform mean vorticity, which induces no velocity on a torus. What remains is the Boussinesq system with a *rotating gravity direction*, in which every wavevector stays an exact grid mode and the buoyancy torque reproduces the transcribed coefficient exactly. A low-frequency force holds the prescribed stratification, which is what the forcing in the theorem being modelled does anyway.

On a background flattened so that its gradient is affine to fourth order at the origin, the full cycle reproduces the reduced model:

	sinusoidal background	flattened background
vorticity landing, $ \Omega(t_b) /\max \Omega $	1.57×10^{-2}	3.67×10^{-4}
steering gain against the reduced model	1.80×10^{-2}	3.06×10^{-6}
hold: drift over three growth times	14.7%	0.36%
endpoint map, worst gap of three pulses	0.196	4.7×10^{-4}

The two columns differ only in the curvature of the background over the layer's support, so that curvature is the entire residual: the reduction assumes an affine background, which the construction arranges by localizing each layer where its background is affine. Both negative controls fail as they must: without the pulse the vorticity is still at 0.99985 of its peak, and with gravity frozen it never turns.

With equal dissipation on both fields the damping enters both amplitude equations on the diagonal, so the whole cycle factorizes as e^{-dt} times the inviscid cycle with $d = \nu\lambda^{2\alpha}$. Three consequences were tested and hold: the *inviscid* pulse amplitude still lands the vorticity at zero (measured 4.1×10^{-4}), a holding interval decays at exactly d (measured to 0.97%), and rescaling the viscous run by e^{dt} reproduces the inviscid one to 0.24%.

3.4 A property of the mechanism that the bookkeeping cannot see [MV]

The first attempt at the cycle, at a small insertion angle, was destroyed by its own background. The stratification the mechanism grows on is Rayleigh-Taylor unstable at rate $\sqrt{A}$, while a layer inserted at angle s grows at $\sqrt{A}\sin s$. Every mode at a more favourable angle therefore gains $1/\sin s$ times as many e-folds as the layer does; over the schedule that run needed, about 72 e-folds from round-off, and the measured amplitude grew by a factor 726 in a holding interval that should have been stationary, with $\max|\theta|$ going from 4.0 to 10.0.

The construction is not vulnerable to this, because each layer is localized and the force controls what happens elsewhere. But it is worth stating plainly: *the cascade always runs against a faster instability of its own background, and what keeps it alive is localization, not speed.*

3.5 What localization costs the dissipative reduction [MV]

Equation (2) is exact for a plane wave, and every layer of the construction is localized, so the damping law is applied to a band of frequencies. Measuring the residual of a compactly supported smooth envelope of radius ℓ on a wave of frequency λ gives a relative error first order in the bandwidth ratio, with slope -0.9863 against a predicted -1 , and depending on that ratio alone: cases sharing $\ell\lambda = 102.4$ but splitting it differently between frequency and envelope agree to four digits. The coefficient is exactly linear in the exponent, 3.606α , which is what the symbol expansion gives, leaving 1.80 as the shape constant of this envelope:

$$\frac{\|(-\Delta)^\alpha(g e^{i\lambda\zeta \cdot x}) - (\lambda|\zeta|)^{2\alpha}g e^{i\lambda\zeta \cdot x}\|}{\|(\lambda|\zeta|)^{2\alpha}g\|} \approx \frac{3.6\alpha}{\ell\lambda}. \quad (3)$$

A control with an envelope as wide as the wave is wrong by 190%, so the measurement can see the failure it tests for. At the separations the published construction uses, $\ell_q^{-1} = \lambda_{q-1}^3$ and $\lambda_q = \lambda_{q-1}^{Q_q}$ with $Q_q \geq 201$, the error is $\lambda_{q-1}^{3-Q_q}$: localization does not obstruct the dissipative reduction and cannot be what sets a hypodissipative threshold.

4 The published threshold, recovered from its own budget

Córdoba, Martínez-Zoroa and Zheng prove blowup for the forced fractional Navier-Stokes equations for every exponent below $\alpha_0 = (22 - 8\sqrt{7})/9$, with a force in $L_t^1 C_x^{1,\epsilon} \cap L_t^\infty L_x^2$. Their introduction gives a back-of-the-envelope value $5 - 2\sqrt{6} = 0.10102$ and says plainly that the real calculation has other error sources. This section identifies which one.

4.1 Four constraints [D]

Every quantity in their construction is a power of one base: the frequency of layer n is $M_n = N^{R^n}$, the amplitude $A_n = N^{aR^n}$, the localization scale $L_n = N^{bR^n}$, and $s > 0$ is the Hölder margin of the force. Their Section 4.3 gives four requirements:

	source	constraint
D dissipation	4.3.5	$a = \alpha R$
S self-interaction	4.3.2	$2a - a/R + b + s - 1 \leq 0$
L localization	4.3.3	$b \geq a + 1/R + s$
O outer velocity on the inner layer	4.3.4	$a + 2/R + s + 1 - 3b \leq 0$

Proposition 4.1 ([D]). *With **D** and **S** saturated, **O** reduces to*

$$4s < 2 + 3\alpha - 7\alpha R - \frac{2}{R},$$

whose maximum over the frequency ratio is at $\alpha R^2 = 2/7$, which is exactly the value the paper chooses. At that optimum $s(\alpha) = (2 + 3\alpha - 2\sqrt{14\alpha})/4$, and $s > 0$ is $9\alpha^2 - 44\alpha + 4 > 0$, whose smaller root is $\alpha_0 = (22 - 8\sqrt{7})/9$.

No coefficient in that chain is fitted; each comes from a displayed exponent. At the published point the localization constraint is *not* active, with margin $\frac{1}{2}(\sqrt{2\alpha/7} - \alpha) = 0.0350$ at α_0 . The introduction's heuristic keeps **D**, **S** and **L** and does not contain **O**; saturating **L** instead and optimizing the same way gives $\alpha R^2 = 1/3$, $s(\alpha) = (1 + \alpha)/2 - \sqrt{3\alpha}$ and the bound $5 - 2\sqrt{6}$. *The entire gap between the heuristic and the proved threshold is a swap of which constraint binds.*

4.2 The admissible frequency ratios [D]

Proposition 4.1 is a quadratic in the frequency ratio, and reading it that way gives a statement we have not seen made explicit.

Proposition 4.2 ([D]). *A positive force margin requires $7\alpha R^2 - (2 + 3\alpha)R + 2 < 0$, so the admissible ratios form the interval with endpoints*

$$R_{\pm} = \frac{(2 + 3\alpha) \pm \sqrt{9\alpha^2 - 44\alpha + 4}}{14\alpha}.$$

The discriminant is exactly the polynomial whose root is α_0 , so the interval closes to a single point precisely at the published threshold, and that point is the paper's own $R = \sqrt{2/(7\alpha)}$. Moreover $R_- > 1$ for every $\alpha > 0$, and at $R = 1$ the margin is exactly $-\alpha$.

α	admissible R	the paper's choice
0.001	(1.002, 285.1)	16.90
0.01	(1.021, 27.98)	5.345
0.05	(1.143, 5.000)	2.390
$\alpha_0 = 0.09267$	$\{1.7559\}$	1.7559
0.10	empty	none

The consequence is structural: a cascade whose frequencies grow geometrically, so that $\log M_n$ is linear in n , is the case $R = 1$, and it is inadmissible at every positive dissipation. Read off the constraints directly, $R = 1$ forces $b \geq a + 1 + s$ from localization against $b \leq 1 - a - s$ from self-interaction, hence $\alpha + s \leq 0$. *Super-geometric frequency growth is forced, not preferred, and more of it is forced as the dissipation rises.*

Proposition 4.1 and Proposition 4.2 are verified in exact arithmetic, by computer algebra, in the public repository's continuous integration.

5 A reduced cascade model, and the retraction of one of its claims

5.1 The model and its threshold [D]

With a geometric background $A_q = A_0 e^{gq}$, a frequency schedule $\lambda_q = c A_q^p$ and amplitudes fixed by the amplification identity, four constraints govern the cascade: summable amplitudes, positive growth at every stage, a finite total time, and survival of each layer's gradient to the blowup time. The time budget turns out not to bind, because stage times decay geometrically, and layer survival coincides with growth positivity, because the remaining time shrinks at exactly the rate the growth rate rises. The threshold is $\alpha_c = 1/(4p)$, measured against a horizon-corrected prediction to 2.59×10^{-6} over 10^6 schedules with zero violations [MV].

5.2 A retraction [D]

An earlier round of this work inverted $\alpha_c = 1/(4p)$ at the published threshold, obtained $p = 11/4 + \sqrt{7}$ exactly, and argued that this was informative because the relation came from constraints not tuned to hit a number and because the resulting p is a clean algebraic number. *Both arguments are withdrawn.* The relation is constraint **D** saturated, and the published family saturates **D** at every α , so $2p = 1/\alpha$ holds identically along it: evaluating at α_0 says nothing about α_0 . And the clean form is automatic, since $\alpha_0 \in \mathbb{Q}(\sqrt{7})$, inverses stay in $\mathbb{Q}(\sqrt{7})$, and the result is tidy only because the norm of $22 - 8\sqrt{7}$ is 36, a perfect square. The retraction is kept here because it is part of the record.

5.3 The cap is unreachable [D]

Allowing any schedule $\log \lambda_{q+1} = R \log \lambda_q$ moves only the amplitude constraint, which becomes $R < p$; the threshold $\alpha_c = 1/(4p)$ stands. Importing nothing but Proposition 4.2 then gives $\alpha_c \leq 1/(4R_-(\alpha))$, and above the published threshold there is no admissible ratio at all. So the cap of $1/4$ that the model measures is correct for the model and vacuous as a bound on the real problem: the schedules that approach it are excluded by the force budget before the dissipation constraint ever binds.

6 Force regularity against dissipation, and a dated prediction

6.1 The exchange rate [D]

The smooth-forcing construction fixes, in its Subsection 3.3, a frequency ratio $\lambda_q = \lambda_{q-1}^{Q_q}$ with $Q_q = Q^* + q$ and $Q^* \geq 200$, a derivative order $k_q = \max\{k : 120k \leq Q_q, \dots\}$, and a seed and gain whose stated consequence is that a layer stops at temperature amplitude $\lambda_q^{-7/8}$. Two facts follow immediately. By the amplification identity the layer deposits a gradient $\lambda_q^{1/8}$, so the *amplitude margin* is $\delta = 1/8$. And the rule $120k_q \leq Q_q$ ties the smoothness of the force to the frequency ratio: each further derivative costs another 120 in the ratio, and a C^∞ force needs $k_q \rightarrow \infty$, which is why Q_q grows with the stage.

Applying (2) stage by stage, growth requires $\nu \lambda_q^{2\alpha} < \sqrt{A_q} \sin s_q$ with $A_q = \lambda_{q-1}^\delta$, hence

$$\boxed{\alpha < \frac{\delta}{4Q_q}} \quad (4)$$

with an insertion angle of order one, and a factor Q_{q-1} tighter with the construction's own angle, which is itself a negative power of the frequency. In the notation of the reduced model this is nothing new: $\lambda_q = A_q^p$ with $p = Q_q/\delta$, and (4) is growth positivity.

Claim 6.1 ([D]). Since $Q_q = Q^* + q$ is unbounded, no positive α satisfies (4) at every stage: a cascade with this schedule and any fixed dissipation runs finitely many stages and stops, and a finite cascade produces a finite gradient.

with the published constants	value
largest α the first stage carries, favourable angle	1.55×10^{-4}
the same with the construction's own insertion angle	7.70×10^{-7}
stages before stalling at $\alpha = 10^{-4}$	113
frequency ratio compatible with $\alpha = 0.0463$	0.675, below one
derivatives of the force affordable there	0.006, none

Read the other way, $120k \leq Q$ with (4) gives $\alpha < \delta/(480k)$, so even at the most generous margin the amplification identity allows, $\delta \rightarrow 1$, controlling a single derivative caps the exponent at $1/480 = 0.00208$: twenty-two times below the threshold already proved for a $C^{1,\epsilon}$ force, and one hundred and seventy-eight times below it at $\delta = 1/8$. The obstruction is therefore not the published constants, which are design choices, but the exchange rate of 120 units of frequency ratio per derivative.

Our own cap of $1/4$ is the corner $\delta \rightarrow 1$, $Q \rightarrow 1$ of the same trade-off; the smooth-forcing design sits at $\delta = 1/8$, $Q \geq 200$. Section 4.2 shows the other corner is closed too, since $R = 1$ is inadmissible.

6.2 The prediction [C]

Claim 6.2 ([C], recorded 2026-09-16). A hypodissipative version of the smooth-forcing mechanism must do at least one of: bound the frequency ratio Q , which by $120k \leq Q$ bounds the derivatives controlled and therefore gives a force of finite regularity rather than C^∞ ; raise the amplitude margin δ toward 1; or change the growth law to one whose rate is the background gradient itself rather than its square root, which is what the vortex-layer construction uses. Its threshold should scale like $\delta/(4Q)$.

The prediction is wrong if a hypodissipative result appears with a positive threshold while keeping a stage-dependent unbounded Q_q , an amplitude margin near $1/8$, and the same square-root growth law. It is also wrong, in a more interesting way, if the dissipative analysis does not damp a layer at $\nu\lambda^{2\alpha}$, which is the one ingredient here that is ours rather than transcribed, and is the ingredient Section 3 tested. The sequel had not appeared when this was recorded, checked against the preprint listings on 2026-09-16 and again on 2026-09-17.

7 Method notes

Every experiment in this program commits its hypothesis and its pass criteria before the run, keeps its raw output, and ends in a verdict that may be CONFIRMED, REFUTED, INCONCLUSIVE or DECIDED IN PART. Two consequences are visible in this paper: a gate that was not met is reported as not met, and the run at the parameters committed for the steering cycle is reported as refuted, with the reason.

Several defects were found by construction rather than by luck, and all are of one family, a check that cannot see what it claims to test: a negative control evaluated where the corruption changed nothing; a slope fitted where the amplitude crosses zero; a horizon too short to see the effect; a wave placed above the dealiasing limit; a measurement window reaching past that limit while the wave itself passed the check; a guard asserting that a background drifts in temperature when the runaway is entirely in the vorticity; and a schedule demanding a phase component of 1.07, which no unit vector has, followed by the reduced model but not by the simulation.

All results reproduce from their own records: each result file stores the settings it was run with, and a harness rebuilds the command line from that block rather than from prose, then diffs every numeric leaf. That distinction is not cosmetic. Rerunning one experiment from its documented command gave a correlation of 0.9920 against a recorded 0.9995, purely because two settings of the recorded run differed from the runner’s defaults, which is indistinguishable from a regression.

8 What is not claimed

This paper does not prove finite-time blowup for any equation; does not verify the mathematics of either announced proof beyond what a kernel replay of a formalization establishes; does not improve any published threshold; and does not claim the layer mechanism, which belongs to Córdoba and Martínez-Zoroa. The threshold reconstruction of Section 4 is exponent bookkeeping with $\delta \rightarrow 0$, logarithms dropped and constants ignored, reproducing the authors’ own stated choices; the trade-off of Section 6 applies a reduction we derived to a schedule designed for an inviscid theorem, which is a statement about what that design costs in a setting it was not built for, not a claim that it fails.

Data and code

Sources, hypotheses, raw outputs, verdicts, the reproduction harness and the exact-arithmetic checks are public at [the project repository](#). The Lean development audited in Section 2 is `openai/NavierStokesAndEuler`; the checker is the `leanchecker` binary shipped with `leanprover/lean4:v4.34.0-rc2`.

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